the value chain

Accelerate. Adapt. Act.

How the SAP Business Technology Platform takes you beyond Robotic Process Automation

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Hyperautomation Expert

Agenda

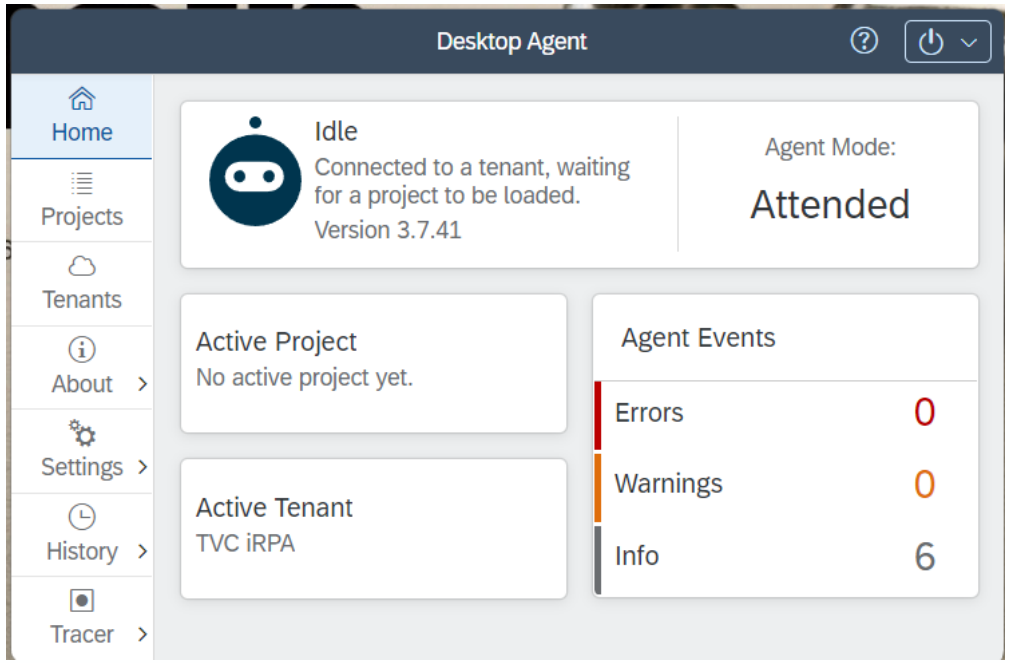
- SAP intelligent Robotic Process Automation & Workflow Management
- SAP Process Automation
- Use Case – Automated Purchase Order Confirmation process

- Demo

SAP intelligent Robotic Process Automation & Workflow Management

SAP intelligent Robotic Process Automation

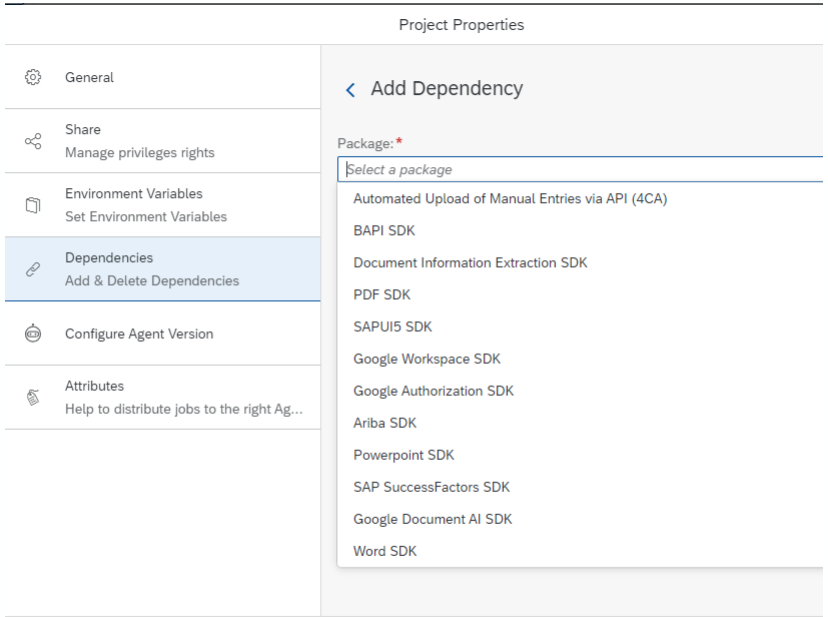
- iRPA (Intelligent Robotic Process Automation)
  - Design, configure, and execute automation projects
    - Design process automations within the Cloud Studio
    - Configure automations in the Environments
    - Execute these packages with Desktop Agents
- Cloud Studio
  - LCNC development environment
- Environments
  - Provide additional details to run automations
- Desktop Agents
  - Locally installed software to run automations
- LCNC: Low Code/No Code (drag and drop activities)



SAP iRPA – standard activities

- Activities – SDK's
  - Outlook, Excel, PDF,… SDK's
  - Document Information Extraction SDK
  - Ariba & SAP SuccessFactors SDK
  - SAP UI5 & SAPGUI SDK
  - BAPI SDK

- New as from release 2209 :
  - Google Workspace SDK
  - Google Authorization SDK
  - Google Vision AI SDK
  - Google Document AI SDK



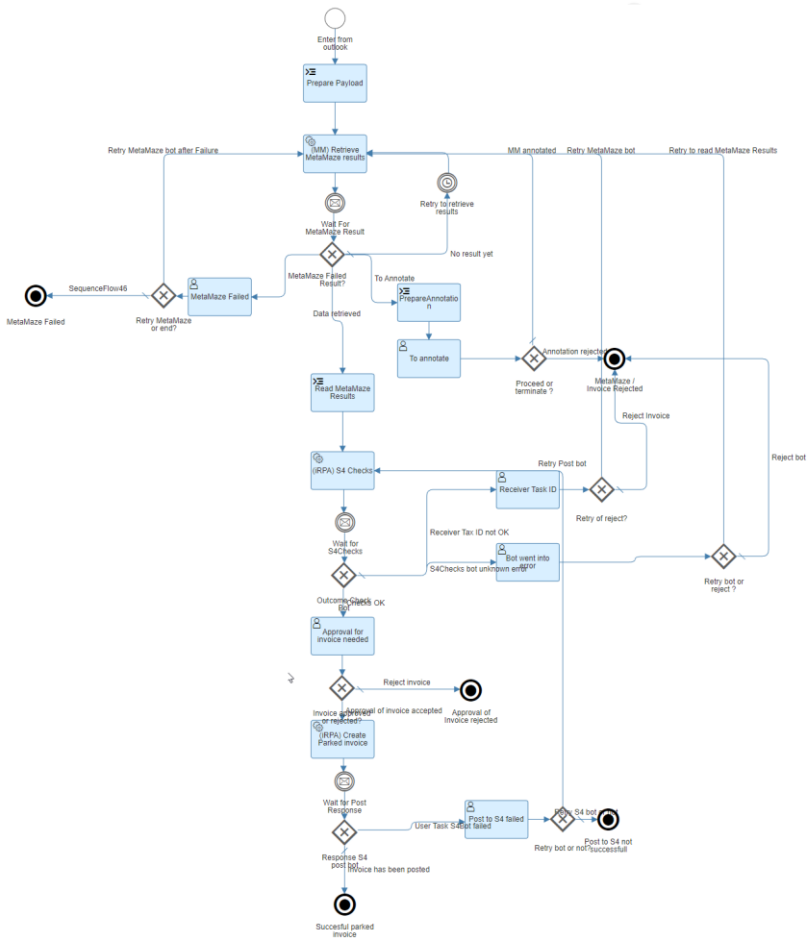
Low Code – ~~No code~~

- Custom scripts still needed
- Retrieving data from Webservice calls
  - Especially error retrieving
- Preparing data for webservice calls
- Preparing BAPI input & reading BAPI responses
- Creating logic and checking data
- Programmer way of thinking
  - Try catch

- Not a single automation without code

Workflow Management

- Used to orchestrate bots, usertasks
- Decision making
  - Human interaction
- Designed in Business Application Studio



Special User Task types

Supplier Material Number Not Matched

Supplier Material Numbers were not matched for PO 4500065206

Created On: Sep 7, 2022
Created By: sb-clone-b7cbe610-c289-4a04-adb8-9bb849a8c761!b130783|workflow!b10150

Priority
Medium

PO Details

PO Number:	4500065206
PDF File:	Confirmation de commande (4500065206)_1662548451488.pdf
Email Subject:	JANVIER LABS - Accusé de réception de votre commande 450006...
Dox Link:	https://process-automation-prd-723rvau2.ui.dox.aiservices.cfapps.e...
List of PDF Items:	Material nr: - Material Supplier: SN-SWISS-M Material nr: - Material Supplier: RN-WI-M
List of SAP Items:	Item nr: 00010 - Material nr: - Material Supplier: CD1JM25-30G Item nr: 00020 - Material nr: - Material Supplier: WI(Han) JM200/225G

Corrected in SAP | Corrected in Dox | Confirmed Manually | Reject | Show Log | Claim

More PDF Lines than PO

Created On: Wed Aug 10 2022 11:28:05 GMT+0200 (Central European Summer Time) READY

Dox ID: fb702a46-6af9-4b10-aa39-4e9c135d3f8c MEDIUM

PO Number: 4500054786

PDF File: AR_C3930033_5134058_1660123356750.pdf

Email Subject: Accuse de reception de commande 4500054786 / 5134058

The PDF contains more Lines than Wanda could find in the PO.
It is necessary to manually correct the results before proceeding
Please proceed through this link
After correcting the results, please approve and continue

Approve | Reject | Show Log | Claim

Commodity Code

Commodity Code has to be filled in for PO 4500054157

The PO was confirmed sucessfully by Wanda, but the Commodity Code still has to be filled in. Please do this manually.

Created On: Aug 10, 2022
Created By: sb-clone-d8e460d9-8410-46bf-a159-cfe1abc80bc0!b129328|workflow!b10150

Priority
Medium

Item Details

PO Number:	4500054157
PDF File:	Sales_Order_Acknowledgement_4244115_1660114149088.pdf
Email Subject:	Sales Order Acknowledgement 4244115
Items & Commodity Codes:	PO Item: 00010 - Commodity Code: 30021200

Commodity Codes Filled in | Show Log | Claim

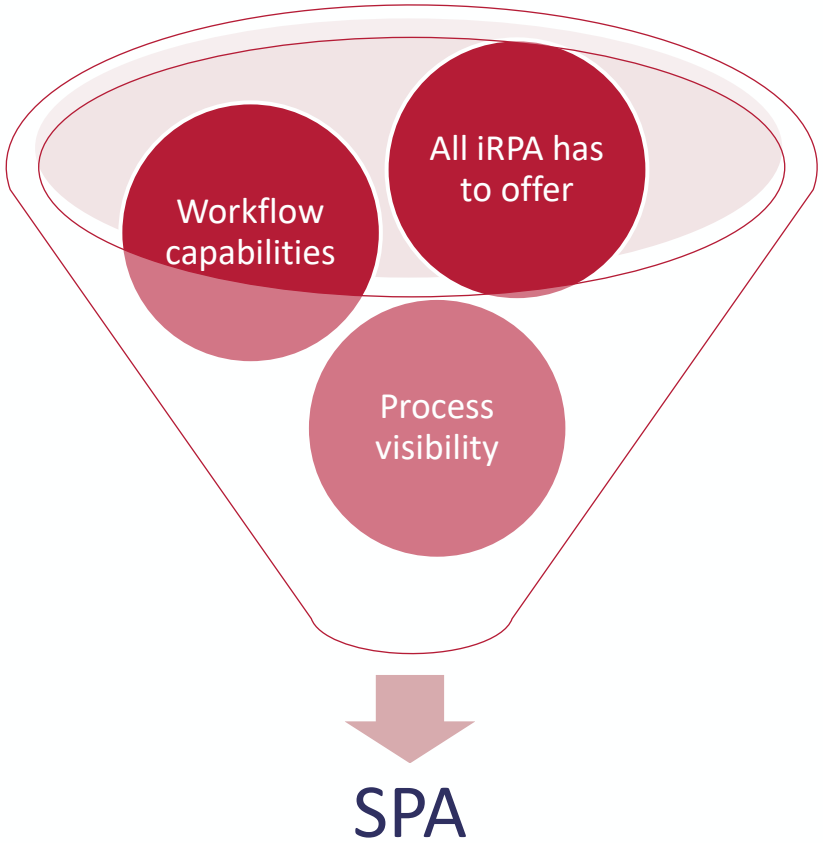
Low Code – ~~No code~~

- Custom scripts needed
- Retrieving data from service tasks
- Sending data to service tasks
- Creating logic and checking data
- Custom UI User Tasks
  - Fiori application

- Not a single automation without code

SAP Process Automation

SAP Process Automation



- Workflow capabilities:
  Link different automation and reach out to end users when needed via the inbox in S4
- Process visibility:
  Create clear metrics so you can follow up on the process.

SAP Process Automation

- Lots of developments ongoing
- Right now
  - iRPA capabilities
  - Business Processes
    - Lightweight Workflow
  - Decision rules
    - Replacement for Business Rules
- Workflow capabilities
  - Roadmap

SAP Process Automation

Q3 2022 CURRENT 24 Road Map Items

Integration and Automation

- Support for Google Workspace (formerly G Suite): productivity suite
  SAP Process Automation

Intelligent Services for Process Automation

- Content recommendations from SAP Signavio Process Insights
  SAP Process Automation
- Integration with Document Information Extraction
  SAP Process Automation

Q4 2022 16 Road Map Items

Future Release

Integration and Automation

- Automatic upgrade of automation agents
  SAP Process Automation
- Remote, desktop–based dense agents
  SAP Process Automation
- Support for Microsoft Office 365 and OneDrive: productivity suite
  SAP Process Automation

Process Automation Content

Q1 2023 15 Road Map Items

Future Release

Integration and Automation

- Agent to support multiple operating system automations
  SAP Process Automation
- Fail-safe automations with resilience to UI upgrades
  SAP Process Automation
- Governance control tower for process automations
  SAP Process Automation
- Integration with SAP One Domain

Q2 2023 6 Road Map Items

Future Release

Integration and Automation

- AI augmenting and assistance for process automation design
  SAP Process Automation
- Self-healing automations based on failure patterns
  SAP Process Automation

Intelligent Services for Process Automation

- AI augmenting and assistance for process automation design
  SAP Process Automation

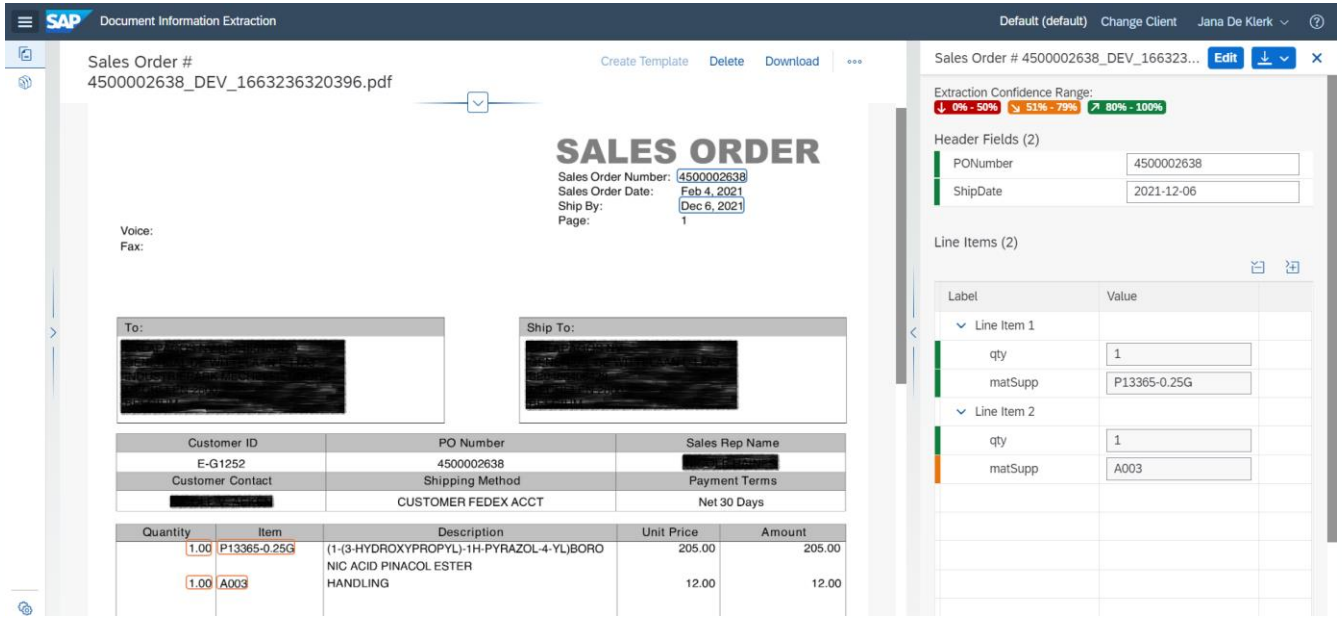
Automation Services

- Intelligent Robotic Process Automation
- Workflow Management
- Process Automation

- Document Information Extraction
- Conversational AI

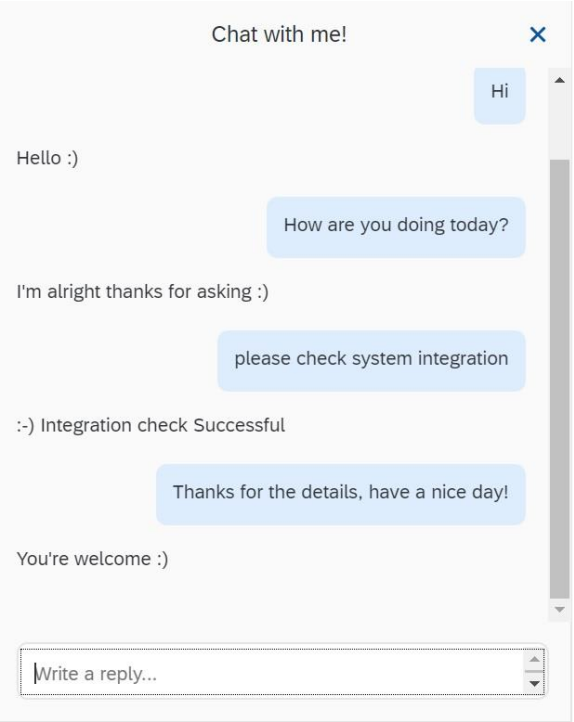
Document Information Extraction

- OCR Service
- Built-in activities in iRPA
  - Easy integration
- Template & schema based reading
- Lots of changes the past month
  - New UI
  - Better reading results
  - New templates
  - Better extractors



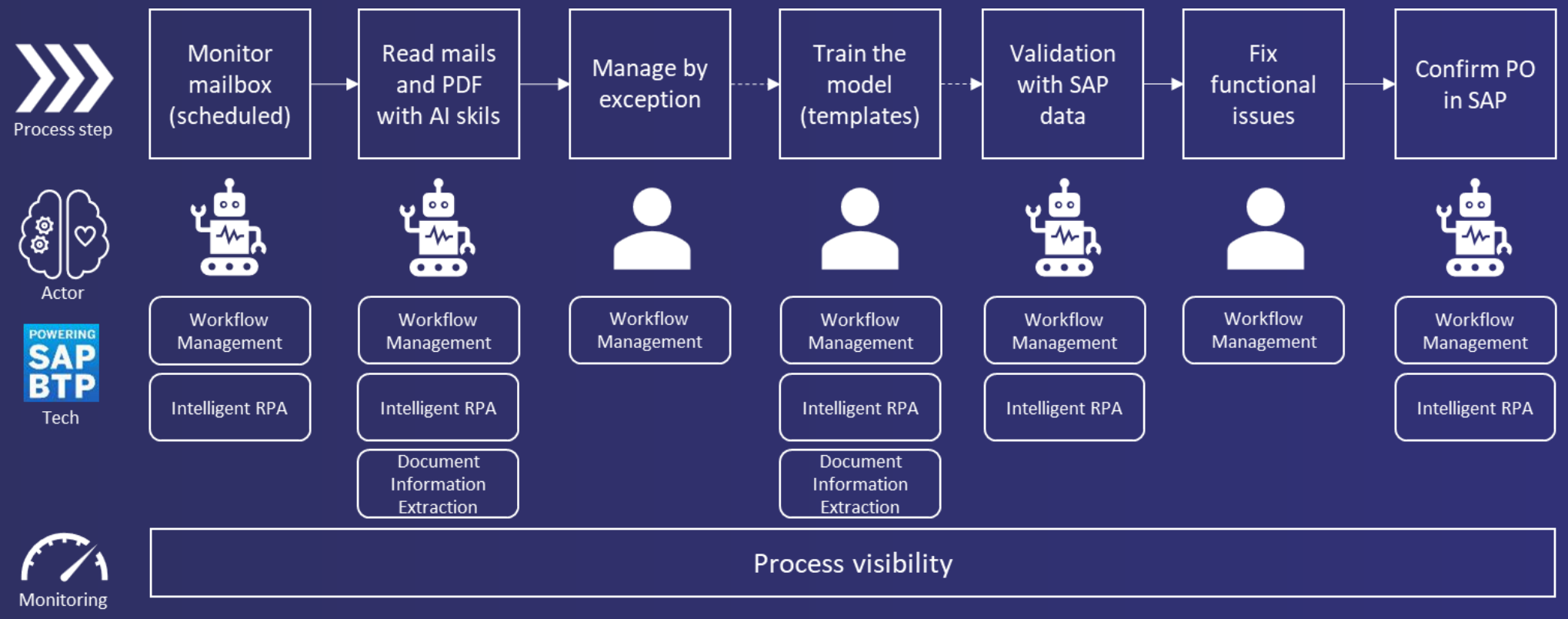
Conversational AI

- Platform to build, train, connect and monitor chat bots
- Chatbot launches iRPA bot
  - Feedback from automation in chatbot
- Integration with MS teams
- Ask the “chatbot” to run a certain process

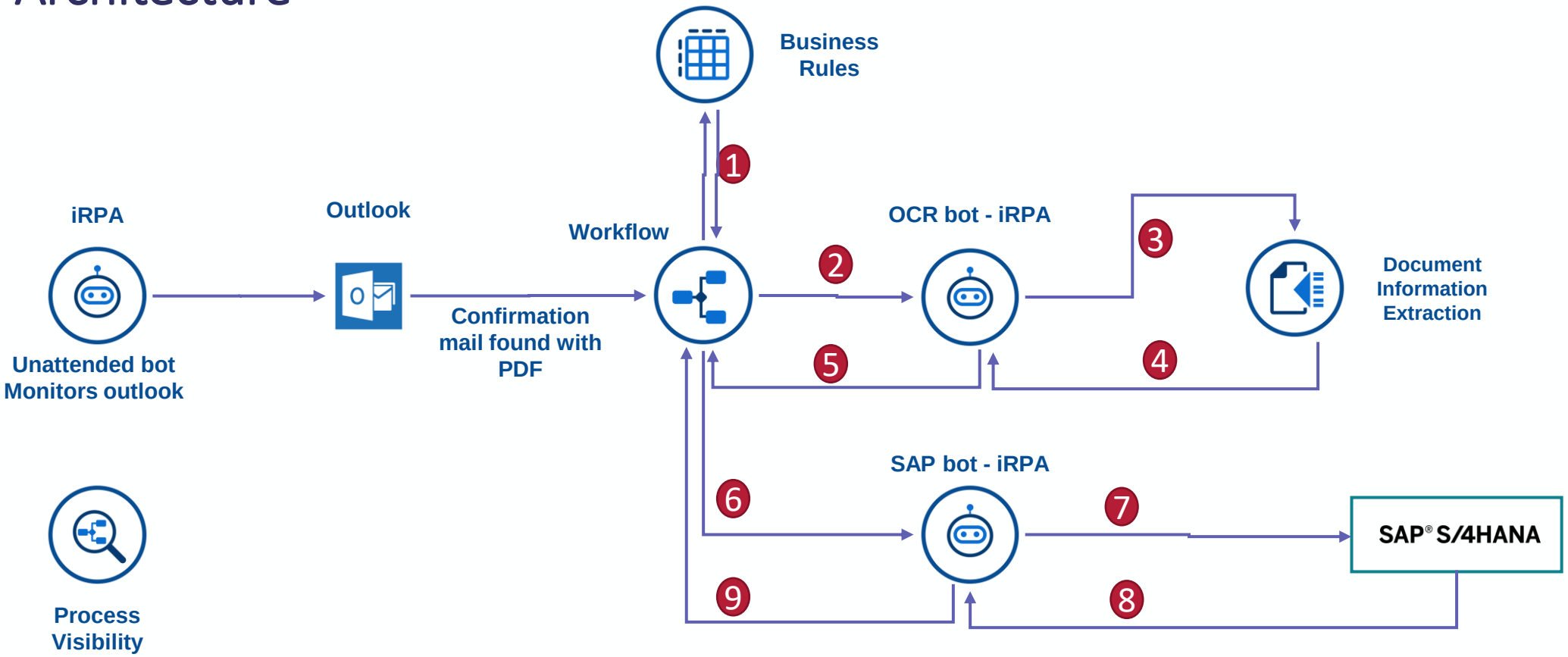


Use case : Automated Purchase Order Confirmation process

Automated workflow for PO confirmations

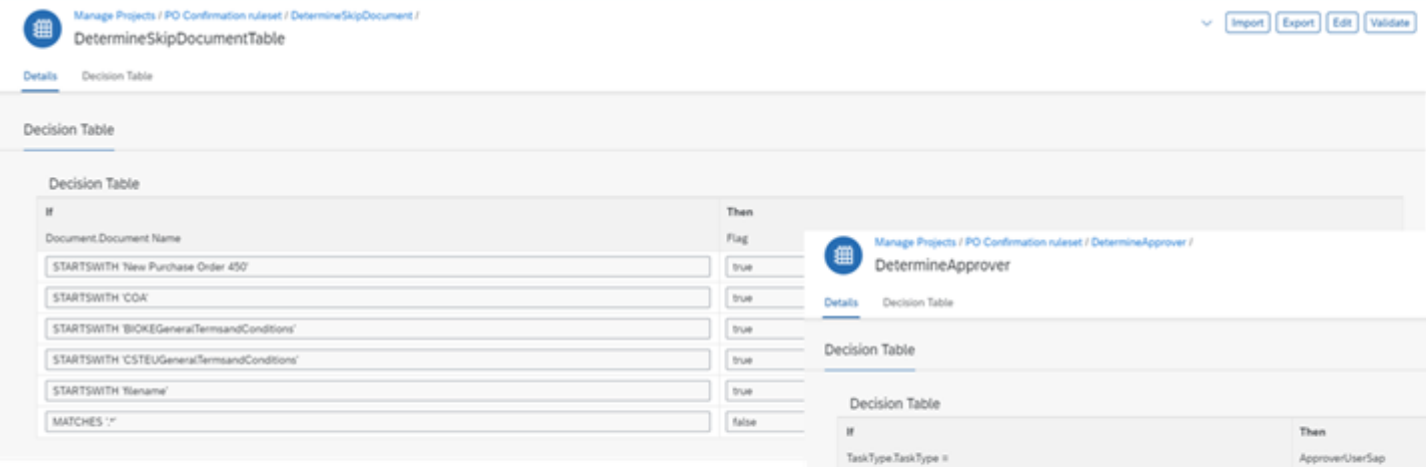


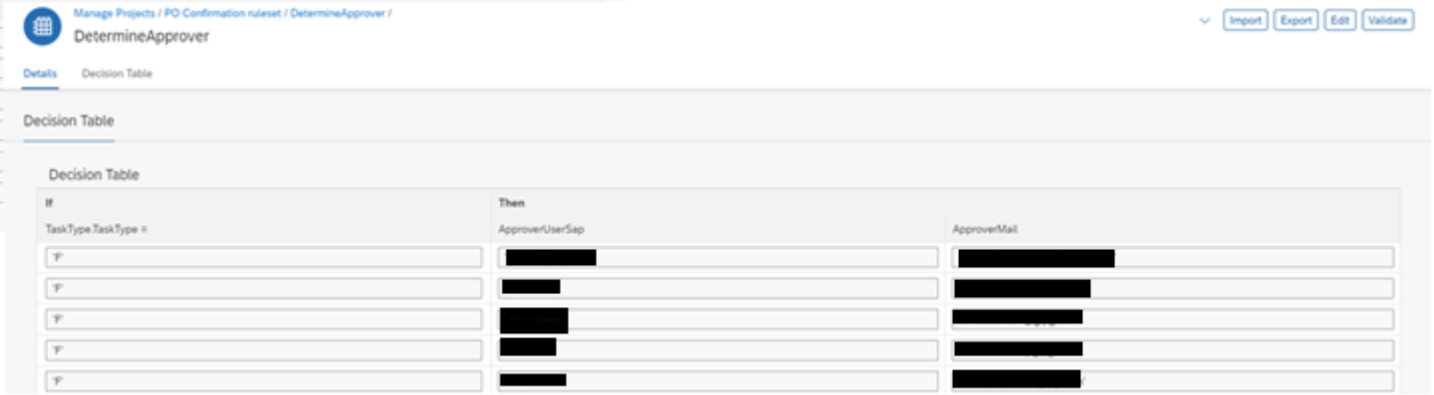
Architecture



Business Rules

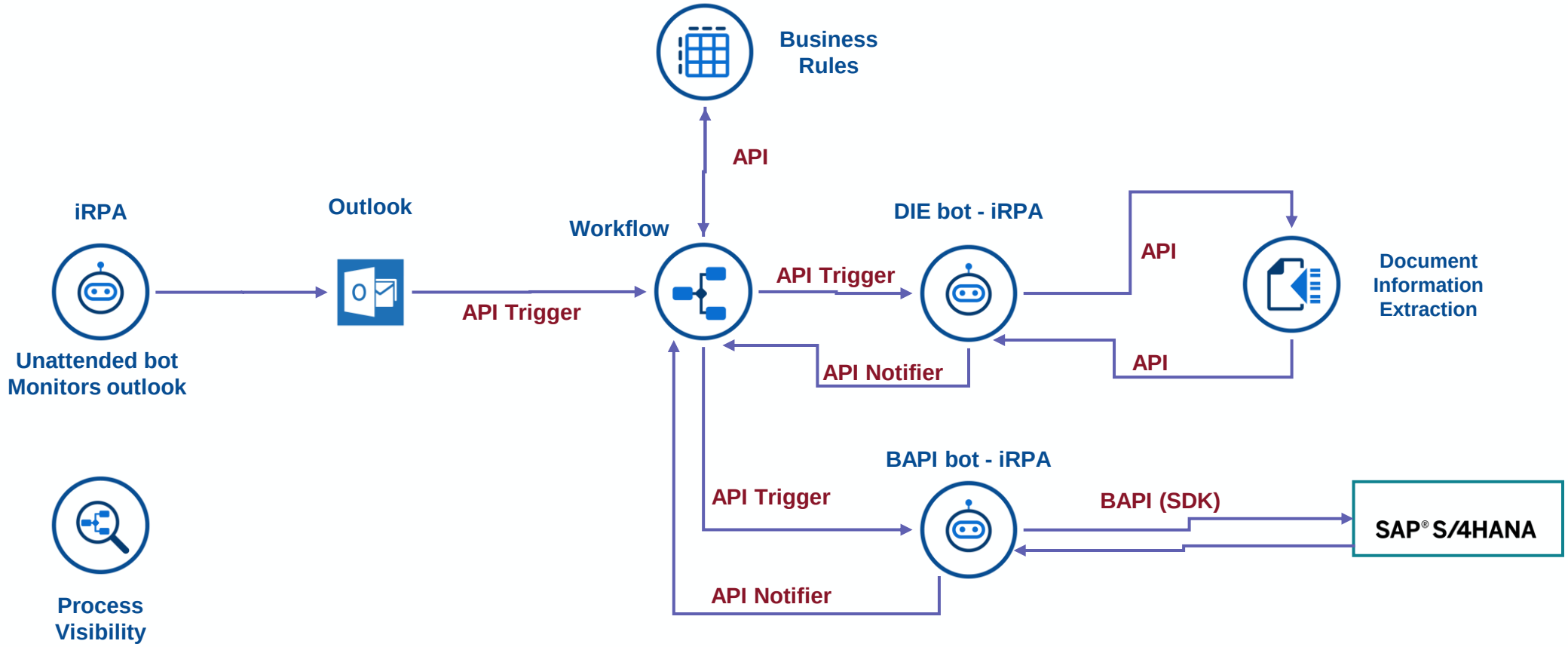
- Business Rules are used by business to change/add logic to the automation





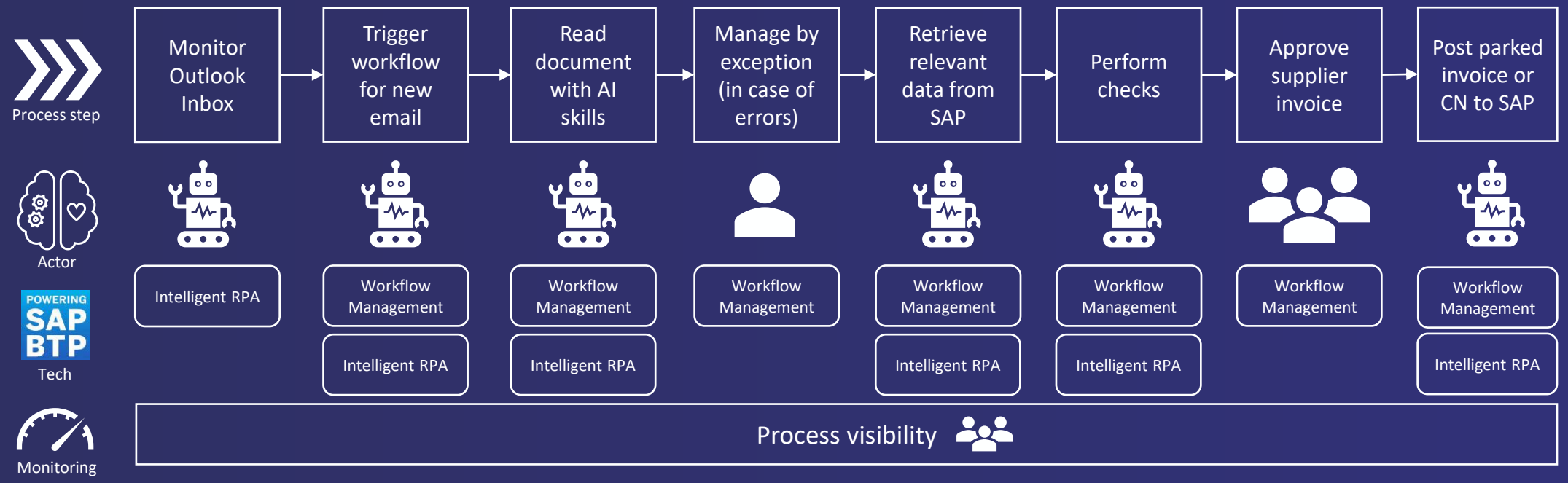
Connecting all the services

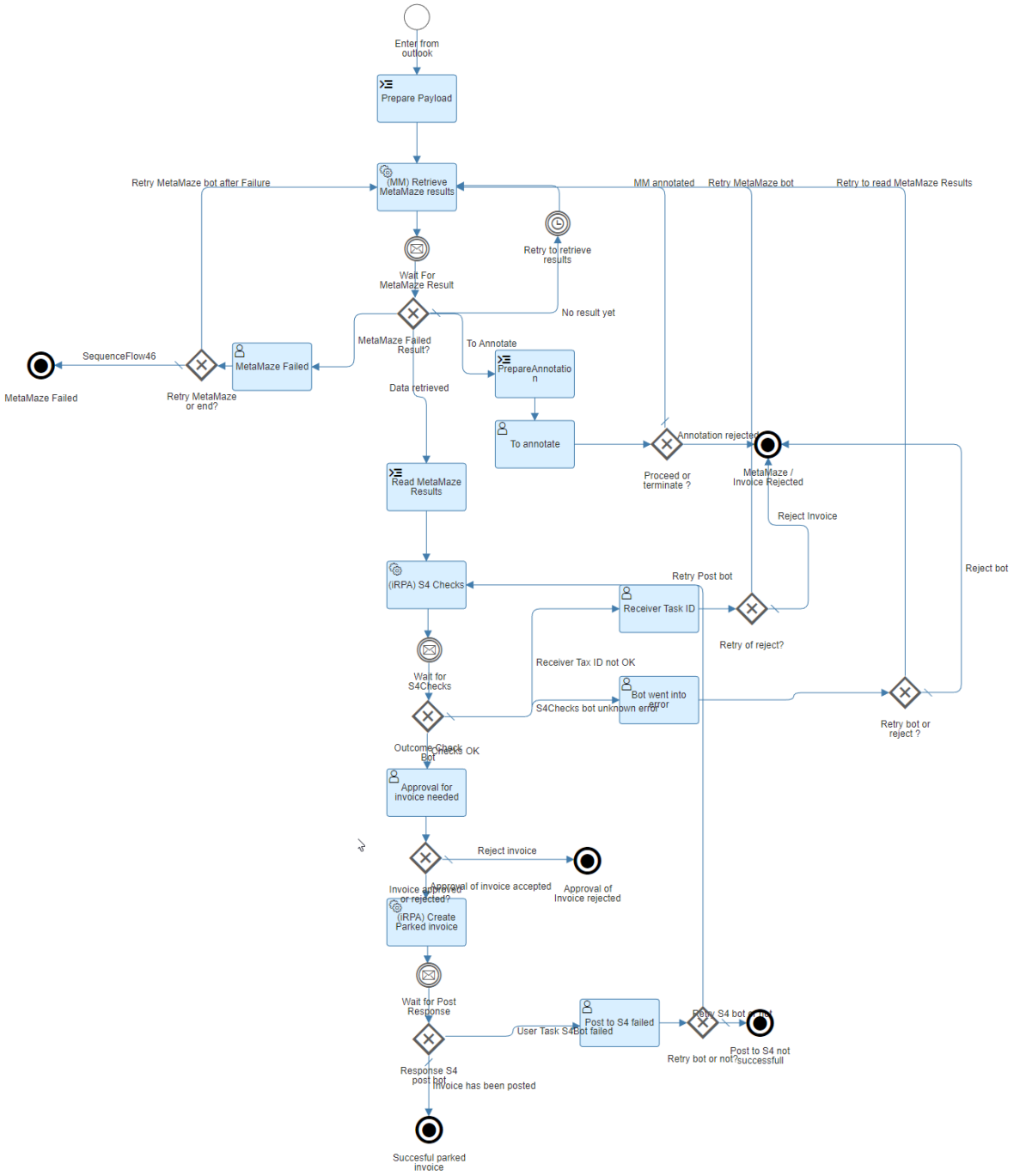
- Via API calls
  - & Notifiers
- BAPI SDK
  - iRPA SDK
  - Allows us to directly call bapi’s
  - Single Sign On / Basic connection


Business Rules
API
iRPA
Outlook
Workflow
DIE bot - iRPA
API
Document Information Extraction
API Trigger
API Trigger
API Notifier
API
Unattended bot Monitors outlook
BAPI bot - iRPA
API Trigger
BAPI (SDK)
SAP® S/4HANA
API Notifier
Process Visibility

Demo – Invoice Scanning Solution

Automated Vendor Invoice Posting




Enter from outlook
Prepare Payload
(MM) Retrieve MetaMaze results
Retry MetaMaze bot after Failure
MM annotated
Retry MetaMaze bot
Retry to read MetaMaze Results
Retry to retrieve results
Wait For MetaMaze Result
No result yet
MetaMaze Failed Result?
To Annotate
Data retrieved
SequenceFlow46
MetaMaze Failed
MetaMaze Failed
Retry MetaMaze or end?
PrepareAnnotation
To annotate
Annotation rejected
Proceed or terminate ?
MetaMaze / Invoice Rejected
Reject Invoice
Reject bot
Read MetaMaze Results
(IRPA) S4 Checks
Retry Post bot
Receiver Task ID
Retry of reject?
Wait for S4Checks
Receiver Tax ID not OK
Bot went into error
S4Checks bot unknown error
Retry bot or reject ?
Outcome Check Bot
Checks OK
Approval for invoice needed
Reject invoice
Approval of invoice accepted
Approval of Invoice rejected
(IRPA) Create Parked invoice
Wait for Post Response
Post to S4 failed
User Task S4Bot failed
Retry S4 bot or not?
Post to S4 not successfull
Response S4 post bot
Invoice has been posted
Succesful parked invoice

Demo

Questions?

Thank you!